Peter Willendrup

Establishing the learning goals, a look at the programme

Learning goals:

1. Learn McStas basics

2. Build and operate simple instrument models, source + optics + sample + detector

3. Learn basics of instrument-optimisation and resolution calculations

5. Add Mantid / NeXus capabilities

6. Get a better idea of what you want to do with McStas, how to do it, how to get help

7. Get up-to-speed with latest developments and advanced features of McStas and related tools like NCrystal and McStasScript.

Enable your independent work with McStas

School programme

November 4th Beginners McStas	Time (GMT)	November 5th Instrument design	Time (GMT)	November 6th Advanced A	November 6th Advan
Welcome + setting learning goals 30 min McStas intro + general concepts 30 min McStas live demo / "Build-along" **Responsible: Peter**	9:00-10:30	30 min Polarisation 60 min Lesser known tips and tricks, including: - Tools for optimisation **Responsibles: Peter**	9:00-10:30	Advanced topics: NCrystal talk Using NCrystal in McStas & McStasscript **Responsible: Thomas**	Work on your own instrument w / help and guidance
Coffee Break	10:30-11:00	Coffee Break	10:30-11:00		Break
60min Guides and gravity: 20 min presentation 40 min practical **Responsible: Mads**	11:00-12:30	Time of flight resolution, talk + exercise **Responsible: Mads**	11:00-12:30	Advanced topics interactive: Write your first component **Responsible: Peter**	Work on your own instrument w / help and guidance
Lunch break	12:30-13:30	Lunch break	12:30-13:30	Lunch break	Lunch break
Choppers / rotating optics: 20 min presentation 25 min practical 45 min Sample + Union overview talk **Responsibles: Mads + Peter**	13:30-15:00	McStas -> Mantid, NeXus: 20 min presentation 20 min demo 50 min exercise **Responsible: Peter**	13:30-15:00	Advanced talk/demo: McStas Union for detectors **Responsible: Mads**	Work on your own instrument w / help and guidance
Coffee Break	15:00-15:30	Coffee Break	15:00-15:30	Coffee Break	Break
45 min Sample exercise 45 min Union exercise **Responsibles: Peter + Mads**	15:30-17:00	90 min McStasScript intro/exercise **Session lead: Mads**	15:30-17:00	Porting old comps and instruments to McStas 3 (and allow GPU execution) **Responsible: Peter**	Work on your own instrument w / help and guidance

Links to various infrastructure and services at:
https://isis2025.mcstas.org

School programme - day 1

November 4th Beginners McStas
Welcome + setting learning goals 30 min McStas intro + general concepts 30 min McStas live demo / "Build-along" **Responsible: Peter**
Coffee Break
60min Guides and gravity: 20 min presentation 40 min practical **Responsible: Mads**
Lunch break
Choppers / rotating optics: 20 min presentation 25 min practical 45 min Sample + Union overview talk **Responsibles: Mads + Peter**
Coffee Break
45 min Sample exercise **45 min Union exercise** **Responsibles: Peter + Mads**

Intro lecture, general principles, sources + monitors,

Lectures + "recipe" exercises

Sample-lecture, including "advanced McStas grammar", sample exercises

In "cookbook" sections, think ahead toward your own project:

* Which neutron source
* What optics
* What sample

- K.I.S.S. for now

School programme - day 2

November 5th Instrument design	
30 min Polarisation 60 min Lesser known tips and tricks, including: - Tools for optimisation **Responsibles: Peter**	Lectures on polarisation and instrument optimisation technicals
Coffee Break	
Time of flight resolution, talk + exercise **Responsible: Mads**	ToF resolution, talk + exercise
Lunch break	
McStas -> Mantid, NeXus: 20 min presentation 20 min demo 50 min exercise **Responsible: Peter**	Mantid-howto, lecture and demo
Coffee Break	
90 min McStasScript intro/exercise **Session lead: Mads**	McStasScript intro/exercise

School programme - day 3, advanced stuff + work on own project

November 6th Advanced A	November 6th Advanced B
Advanced topics: NCrystal talk Using NCrystal in McStas & McStasscript **Responsible: Thomas**	Work on your own instrument w / help and guidance
	Break
Advanced topics interactive: Write your first component **Responsible: Peter**	Work on your own instrument w / help and guidance
Lunch break	Lunch break
Advanced talk/demo: McStas Union for detectors **Responsible: Mads**	Work on your own instrument w / help and guidance
Coffee Break	Break
Porting old comps and instruments to McStas 3 (and allow GPU execution) **Responsible: Peter**	Work on your own instrument w / help and guidance

For the exercise-based work-sessions

- You will benefit from working in pairs,   2 > 1

- Take turns being the “coder”  and the “parallel processor” 

Let's get to it!

2025 ISIS McStas School